

Austin Lin

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Open to Relocation

TECHNICAL SKILLS

Languages: Python, Java

Tools: MATLAB/Simulink, ROS2/Gazebo

EXPERIENCE

General Dynamics Electric Boat

Electrical Systems Engineer, New London, CT

Sept 2024 - Present

- Implemented new shock modeling method of electrical cables, reducing excess cable length/mass by up to 60% compared to existing methods, in a space-constrained environment. Used MATLAB to implement statistical methods and cost-constraint optimization to fuse data from specifications, finite element shock studies, and other sources.
- Led a team of engineers to implement class-wide design changes to improve safety and reliability.
- Bypassed production bottlenecks by providing engineering assistance to other disciplines (Design, Work Planning, Deckplate/Trades), supporting ongoing schedule acceleration efforts.

L3Harris C5 Systems

Systems Engineering Intern, Camden, NJ

June 2023 – Aug 2023

- Demonstrated product quality to the customer by performing System Operability Verification Tests for Marcom IVCS and Symphony. Completed CompTIA Network+ Training.
- Performed root-cause troubleshooting on network/radio configurations with Linux-based hardware.

UMD Space Systems Lab

Undergrad Researcher, College Park, MD

May 2022 – Aug 2022

- Delivered spacecraft robot arm testing and modeling results to facilitate In-Space Servicing mission requirements. Conducted research with Benjamin Reed from Quantum Space and Dr. David Akin.
- Calculated and visualized workspace of a robotic arm to reach payloads for in-orbit servicing. Used Denavit-Hartenberg parameters to calculate forward kinematics.
- Designed and fabricated a 1/4-scale physical robotic arm prototype. Produced and executed a test procedure to validate Python model results for 24+ spacecraft configurations.

PROJECTS

Battery Simulation and State of Charge Estimator

- Created a Python/Shiny app to visualize the impact of adjusting Luenberger observer gain and measurement noise on battery state of charge estimation accuracy and convergence.

Robotics and Autonomous Systems

- Implemented a maze-solving algorithm into a robot using LIDAR for live localization and mapping. Used Gazebo and ROS2 to model a virtual environment to tune the algorithm before field testing.
- Programmed robots to identify, grab, and hand off Legos to each other using computer vision. Trained a YOLOv8 model for object identification and odometry, and used ZeroMQ through TCP/IP for communication.
- Developed hardware and software for a robot arm and controller. Programmed C++ firmware for an Arduino microprocessor to handle serial communication to a computer that calculates and graphically displays arm kinematics.

EDUCATION

University of Maryland, College Park, MD

Aug 2020 – Aug 2024

B.S. Mechanical Engineering, Minor in Robotics and Autonomous Systems

Scholars Public Leadership Program